Naveen K

AI/ML Engineer & Researcher

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Profile

AI/ML Engineer actively building end-to-end automation pipelines in production and conducting research in AGI, ASI, and OI systems. Experienced in LLM fine-tuning, RAG system development, Generative AI, OCR-based intelligent document processing, Named Entity Recognition (NER), and Deepfake Detection using TensorFlow and PyTorch. Skilled in designing and deploying full-stack AI pipelines from data ingestion to production deployment using Python and REST APIs. Passionate about pushing the boundaries of AI toward general and superintelligent systems.

Professional Experience

Junior AI Engineer, *ATna.ai* — 02/2026 – 05/2026, Chennai

- Built production-ready intelligent document processing pipelines using OCR, NER, and Regex, reducing manual data entry effort by 60%.
- Trained and fine-tuned NLP/NER models for extracting phone numbers, emails, addresses, and business entities from 1,000+ documents.
- Improved entity accuracy by 25% using custom regex validation
- Integrated OCR engines with Generative AI workflows processing 500+ documents daily in production.

AI/ML Intern, *Technical One* — 01/2026 – 02/2026, Remote

- Built a Fake News Detection system using NLP and ML classification models, achieving reliable accuracy on real world news article datasets.
- Designed end-to-end text preprocessing pipeline including tokenization, stopword removal, and feature extraction for supervised learning tasks.
- Optimized model performance through hyperparameter tuning and cross-validation, improving detection.

Data Science Intern, *VCodez Infotech* — 05/2025 – 08/2025, Chennai

- Collected, cleaned, and analyzed datasets to extract meaningful insights.
- Performed exploratory data analysis (EDA) and created visualizations using Python.
- Built and evaluated machine learning models for predictive analysis.

Education

IIT-M Advanced Programming Professional & Master Data Science, *IIT-M GUVI* — 2024 – 2025, Chennai

Bachelor of Business Administration (BBA), *H-K-R-H College* — 03/2021 – 06/2024, Theni

Skills

- Python
- LLM Fine-tuning
- TensorFlow / PyTorch
- NER
- NumPy
- Machine Learning
- LangChain
- Qdrant (Vector Database)
- openCV
- Git/GitHub
- Deep Learning
- RAG (Retrieval-Augmented Generation)
- REST API Development
- Computer Vision
- scikit-learn
- Generative AI
- Hugging Face / Transformers
- OCR
- Pandas
- Docker

Projects

Navix AI & AGI Developer Platform, *Tech Stack: Python, FastAPI, Qdrant, Ollama (Llama 3), LangChain, Crew AI, MCP, GraphRAG, Hierarchical RAG, HNSW, Faster-Whisper, Docker, VLLM, SGLang, LLMOps, Quantization, Model Merging, Continual Learning, Multimodal AI, Pydantic, SQLAlchemy, AES-GCM, SSE Streaming, Custom Hardware* — 04/2026 – Present

- Built a production-grade offline AGI platform with switchable online mode, zero cloud dependency, and custom hardware infrastructure.
- Designed an advanced RAG pipeline using GraphRAG, HNSW indexing, Qdrant vector search, and hierarchical retrieval for large-scale document processing.
- Developed a multi-agent orchestration system using Crew AI and MCP with LLM fine-tuning, quantization (INT4/INT8), model merging, and VLLM/SGLang inference optimization.
- Engineered a multimodal pipeline supporting text, voice, PDF, DOCX, CSV and image inputs with self-healing DevOps daemon and LLMOps practices.

Deepfake Detection System, *Key Skill : Python, Deep Learning, CNN, OpenCV, TensorFlow/PyTorch, Computer Vision* — 03/2026 – 05/2026

- Built an AI-powered Deepfake Detection system achieving 94% classification accuracy on real and manipulated media datasets.
- Trained CNN-based models to detect facial inconsistencies and synthetic media patterns, improving detection reliability by 28%.
- Processed 10,000+ video frames using OpenCV for frame extraction, facial preprocessing, and image enhancement.
- Trained on 15,000+ real and fake media samples for binary classification with augmentation pipelines reducing overfitting by 22%.
- Reduced false positives by 30% and optimized inference speed for production-level deployment.
- Improved model performance through hyperparameter tuning, reducing validation loss by 18%.

Certificates

- Master Data Science : Guvi
- Data Science Intern : VCodez Infotech
- AIML intern - Technical One
- Junior AI engineer - Atna.ai